# Lucas S. Krippendorff

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## Education

**University of Pennsylvania** Aug 2024 – May 2028
*B.S.E. Electrical Engineering, GPA 3.51* *Philadelphia, PA*
- **Coursework:** Embedded Systems, Digital Circuit Design, Computer Systems; *Grad:* Digital IC & VLSI, Networks & Protocols

## Experience

**SubVysion (YC S26)** Mar 2026 – Present
*Hardware Engineer Intern · part-time during academic year, full-time Summer 2026* *Philadelphia, PA*
- First engineering hire and hardware lead for an autonomous outdoor GPR mapping rover; designed and built the first rover in under two weeks, replacing a manual sensor cart
- Integrated the sensor and compute stack on an NVIDIA Jetson Orin NX, including three ZED X stereo cameras over GMSL2, RTK GNSS, IMU, ground-penetrating radar, and a 5G uplink
- Evaluated and down-selected compute, GNSS, and LiDAR hardware across rover revisions, iterating from Jetson Xavier NX to Orin Nano to Orin NX and comparing Livox Mid-360 and Unitree L2 LiDARs
- Architected 36 V LiFePO4 power distribution with an anti-spark main switch, a 240 W buck converter feeding a fused 12 V bus, and brake resistors sized for ODrive regen
- Wired CAN control of dual ODrive S1 BLDC drives with vibration-resistant JST-GH, GX12, and M8 connectors
- Led mechanical CAD in Onshape and packaged electronics in non-metallic G10/FR4 and ABS enclosures for GPR compatibility; authored the BOM and assembly documentation for future builds

**3D Chimera** Jun 2025 – Jul 2025
*Engineering Intern · 3D-printing solutions & Bambu Lab reseller* *Miami, FL*
- Engineered a hybrid storage system pairing off-the-shelf materials with custom 3D-printed parts to cut cost and assembly time
- Repaired and calibrated Bambu Lab X1E and H2D printers; prototyped parts with tuned infill and material profiles, testing tensile, shear, and sealing performance

## Projects

**Six-DOF Robotic Arm** | *Personal Project* May 2025 – Aug 2025
- Designed and built custom 26:1 cycloidal and 5:1 planetary gearboxes with integrated bearings on NEMA-17 steppers
- Built an inverse-kinematics solver and trajectory planner in Python, streaming step pulses to C++ Arduino firmware, with a JavaScript/Meshcat interface for real-time waypoint control

**USB-C Speaker & Discrete Loudness Meter** | *Personal Project* Aug 2026 – Present
- Designed a USB-C amp board in KiCad that enumerates as a USB sound card (PCM2704), negotiates 12 V over PD (CH221K), and drives a TDA7297 class-AB stage, iterating through 4 JLCPCB-assembled revisions
- Built a 20-LED stereo loudness meter from discrete MMBT3904 stages as a flash-ADC-style ladder, sizing each base divider by hand so thresholds fall at equal dB steps, snapped to E24 values

**22nm CMOS Datapath & Memory** | *UPenn ESE 3700* Jan 2026 – May 2026
- Designed an 8-bit ripple-carry adder and a synchronous 16×4 6T SRAM at the transistor level (22nm HP PTM, 0.8 V)
- Verified SRAM reads and writes in SPICE up to 3.36 GHz (298 ps minimum period), 6.7× above the 500 MHz spec
- Cut worst-case carry delay 27% using a transmission-gate XOR2 over a NAND baseline, checked with Elmore and SPICE

**Cat–Dog CNN Classifier** | *UPenn ENGR 1050* Dec 2024
- Trained a PyTorch CNN on 24,000 images with CUDA acceleration to 97.8% test accuracy, building the full preprocessing pipeline

## Activities & Skills

**Activities:** UPenn Mars Rover Club (2025–Present), UPenn Aerial Robotics Team (2024–25)
**Programming:** Python (PyTorch, CUDA), C/C++ (embedded firmware), JavaScript, Git
**Hardware:** NVIDIA Jetson, STM32 / ESP32 / ATmega, BLDC & stepper control (ODrive), CAN / UART / I2C / SPI, RTK GNSS, stereo cameras, GPR, LiFePO4 / LiPo systems, buck converters, PCB design, SPICE
**Fabrication & CAD:** Onshape, SolidWorks, Fusion 360, KiCad, Electric VLSI, 3D printing, G10/FR4 composites, wiring harnesses
**Languages:** English (native), Japanese (beginner)